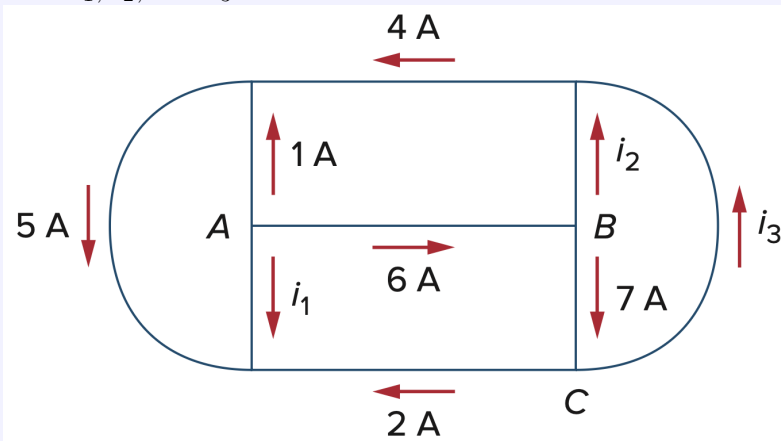


Exercise Sheet #2: DC 1: Circuit Calculations

Discussion: Oct 22, 2025

Exercise 1: KCL

Find i_1 , i_2 , and i_3 .

Solution

$$-1 \text{ A} - 6 \text{ A} - i_1 = 0$$

$$\Rightarrow i_1 = -1 \text{ A} - 6 \text{ A} = -7 \text{ A}$$

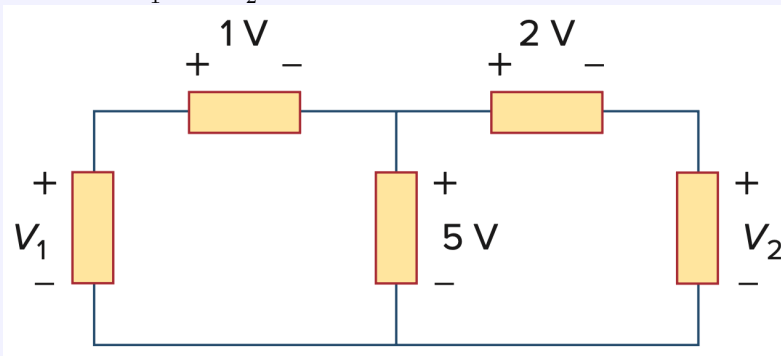
$$7 \text{ A} - 2 \text{ A} = 5 \text{ A} - i_3 = 0$$

$$\Rightarrow i_3 = 7 \text{ A} - 2 \text{ A} = 5 \text{ A}$$

$$i_2 + i_3 - 4 \text{ A} = 0$$

$$\Rightarrow i_2 = 4 \text{ A} - i_3 = -1 \text{ A}$$

Exercise 2: KVL

Calculate V_1 and V_2 .

Solution

$$1\text{ V} + 5\text{ V} - V_1 = 0$$

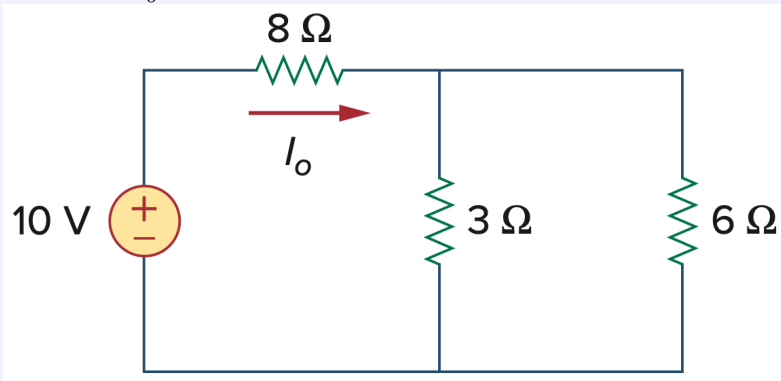
$$\Rightarrow V_1 = 1\text{ V} + 5\text{ V} = 6\text{ V}$$

$$2\text{ A} + V_2 - 5\text{ V} = 0$$

$$\Rightarrow V_2 = 5\text{ V} - 2\text{ V} = 3\text{ V}$$

Exercise 3: Series and Parallel Resistors

Calculate I_0 .

**Solution**

First, we calculate the equivalent resistance of the parallel resistors with 3Ω and 6Ω , then we add the series resistor with 8Ω :

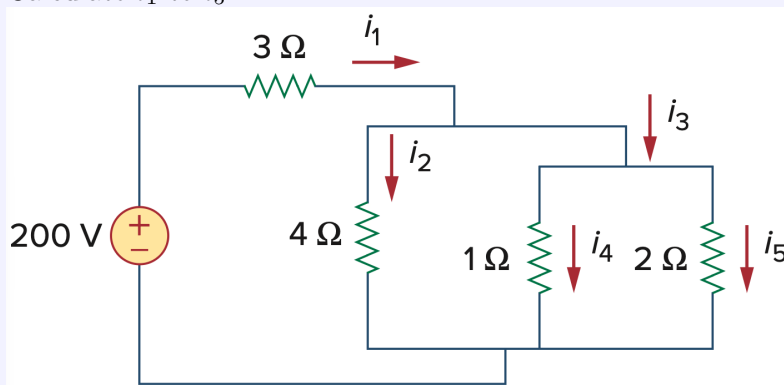
$$R = \frac{3\Omega \cdot 6\Omega}{3\Omega + 6\Omega} + 8\Omega = 10\Omega$$

Finally, we can calculate the current I_0 using Ohm's law:

$$I_0 = \frac{10\text{ V}}{10\Omega} = 1\text{ A}$$

Exercise 4: Current Divider Rule

Calculate i_1 to i_5 .



Solution

First, we calculate the equivalent resistance of the parallel resistors with $1\ \Omega$ and $2\ \Omega$ and call it R_3 :

$$R_3 = \frac{1\ \Omega \cdot 2\ \Omega}{1\ \Omega + 2\ \Omega} = \frac{2}{3}\ \Omega$$

Then we calculate the total resistance R from the $4\ \Omega$ resistor parallel to R_3 and the $3\ \Omega$ resistor in series:

$$R = 3\ \Omega + \frac{4\ \Omega \cdot \frac{2}{3}\ \Omega}{4\ \Omega + \frac{2}{3}\ \Omega} = \frac{25}{7}\ \Omega$$

Using Ohm's law, we can calculate the total current i_1 :

$$i_1 = \frac{200\ \text{V}}{\frac{25}{7}\ \Omega} = 56\ \text{A}$$

Then, we can calculate i_2 using current division:

$$i_2 = i_1 \cdot \frac{R_3}{R_2 + R_3} = 56\ \text{A} \cdot \frac{\frac{2}{3}\ \Omega}{4\ \Omega + \frac{2}{3}\ \Omega} = 8\ \text{A}$$

KCL gives us i_3 :

$$i_3 = i_1 - i_2 = 56\ \text{A} - 8\ \text{A} = 48\ \text{A}$$

Again, we can calculate i_4 using current division:

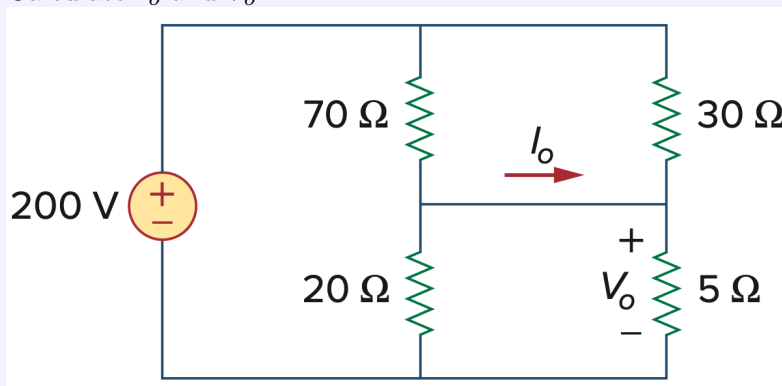
$$i_4 = i_3 \cdot \frac{R_5}{R_4 + R_5} = 48\ \text{A} \cdot \frac{2\ \Omega}{1\ \Omega + 2\ \Omega} = 32\ \text{A}$$

Finally, KCL gives us i_5 :

$$i_5 = i_3 - i_4 = 48\ \text{A} - 32\ \text{A} = 16\ \text{A}$$

Exercise 5: Node Connection

Calculate I_o and V_o .



Solution

A node is a point where two or more circuit elements connect (electrically the same point). A branch is a path that connects two nodes and contains a circuit element (resistor, source, etc.).

Here, there are three main nodes:

- Top node: connected to the +200 V source, top of the 70 Ω and 30 Ω resistors.
- Bottom node: 0 V reference.
- Middle node: connection between two branches. On the left, the branch formed by the 70 Ω and 20 Ω . And on the right, the branch formed by the 30 Ω and 5 Ω .

The middle horizontal wire (with I_o) is part of the middle node, but because there's a potential difference between the left and right parts of that node before it's connected, current flows through that connection — hence we analyze I_o as if it's a branch current.

As the output voltage V_o is the voltage across the 5 Ω resistor, the output voltage equals the middle node voltage.

Currents from the top node to the middle node:

$$I_{70} = \frac{200 \text{ V} - V_o}{70 \Omega}, \quad I_{30} = \frac{200 \text{ V} - V_o}{30 \Omega}$$

Currents from the middle node to the bottom node:

$$I_{20} = \frac{V_o}{20 \Omega}, \quad I_5 = \frac{V_o}{5 \Omega}$$

Applying KCL at the middle node:

$$\frac{200 \text{ V} - V_o}{70 \Omega} + \frac{200 \text{ V} - V_o}{30 \Omega} = \frac{V_o}{20 \Omega} + \frac{V_o}{5 \Omega}$$

Simplify:

$$\frac{200 \text{ V} - V_o}{21 \Omega} = \frac{V_o}{4 \Omega}$$

$$\Rightarrow 800 \text{ V} - 4 \cdot V_o = 21 \cdot V_o$$

$$\Rightarrow \boxed{V_o = 32 \text{ V}}$$

Now we use KCL to compute the current I_o (defined to the right):

$$I_o = I_{70} - I_{20} = \frac{200 \text{ V} - 32 \text{ V}}{70 \Omega} - \frac{32 \text{ V}}{20 \Omega} = \boxed{0.8 \text{ A}}$$

Exercise 6: ...

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Solution

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Exercise 7: ...

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Solution

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Exercise 8: ...

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Solution

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Exercise 9: ...

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Solution

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